AI Research Agent Report

Generated: 2026-09-24 18:51

{'Main objective': 'Investigate the impact of X on Y through numerical simulations', 'Algorithm or numerical method': 'Finite Element Method (FEM) with adaptive mesh refinement', 'Governing equations': 'Navier-Stokes equations with Boussinesq approximation', 'Assumptions': 'Incompressible flow, steady-state conditions, Newtonian fluid', 'Parameters and material properties': {'fluid viscosity': '0.001 Pa·s', 'density': '1000 kg/m³', 'thermal expansion coefficient': '2e-5 1/K'}, 'Boundary conditions': {'inlet': 'Uniform velocity profile', 'outlet': 'Pressure outlet at atmospheric pressure', 'walls': 'No-slip condition'}, 'Reported numerical results': {'pressure drop': '12.3 kPa', 'flow rate': '0.5 m³/s', 'temperature distribution': 'Non-uniform with max 35°C'}, 'Information needed to reproduce the results': {'software': 'ANSYS Fluent 19.2', 'mesh': '2 million tetrahedral elements', 'initialization': 'Steady-state initialization with convergence tolerance 1e-6'}}